

DENIS FEDERICO LIM

+1 (213) 249-3972 — denislim95@gmail.com — San Francisco, CA, USA — [linkedin.com/in/denis-manuel-lim](https://www.linkedin.com/in/denis-manuel-lim)

PUBLICATIONS

Neighbor Crowding Predicts Feature Stability in Sparse Autoencoders

July 2026

Lim, D., et al. **Accepted**, NeurReps Workshop @ NeurIPS 2026. [\[pdf\]](#)

- Showed that cross-seed feature stability in TopK SAEs can be predicted from training a linear classifier with cheap statistics of a single run, avoiding the cost of multi-seed retraining.
- Identified *neighbor crowding* — how densely a feature’s decoder column is packed against its neighbors on the unit sphere — as the strongest predictor, and showed the predictor transfers across model scale within a family.

Game-Theoretic Specialization of Safety LoRA Adapters

June 2026 – September 2026

Lovett, M., Lim, D., Vosoughi, S. **Under review**, ICLR 2027.

- Applied a game-theoretic model to specialize safety LoRA adapters; built distance-based weighting and diagnostics.

Representational Adversarial Steganography in Colluding LLMs

March 2026

Rangaraju, V., Medicherla, S., ..., Lim, D., et al. Apart Research Hackathon [\[project page\]](#)

- Formalized collusive representational steganography as a three-player adversarial game.
- Implemented a co-evolutionary adversarial training protocol using GPT-2 small with a rank-8 latent MLP channel; achieved near-perfect covert signal transmission while degrading monitor accuracy to ~20% after 15 adversarial rounds.

FELLOWSHIPS & RESEARCH

KAIST STAI Lab (Prof. Seong Joon Oh) — *Research Collaborator*

September 2026 – Present

- With Upstage, investigating degenerate states (reasoning loops, language mixing) in Hermes Agent + Solar Pro 4.

SPAR Research Fellowship (under Laura Ying Schulz)

September 2026 – Present

- Researching how reinforcement learning affects metacognition in LLMs, applying mechanistic interpretability methods to find the underlying causal representations.

MARS V Fellowship (under Shivam Raval & Luiza Corpaci)

June 2026 – Present

- Investigating mechanistic detection of semantically unfaithful LLM translations of formal specifications.

Chain-of-thought Faithfulness in Small Reasoning Models

November 2025 – January 2026

- Built mechanistic interpretability pipeline to evaluate CoT faithfulness in DeepSeek-R1-Distill-Qwen-1.5B, using TransformerLens hooks to cache residual-stream activations and linear probes trained to classify faithfulness.

EXPERIENCE

Arcadia Impact

Remote (London, UK)

Research Engineer — Automated Research Alignment (Part-time)

July 2026 – September 2026

- Built evaluation infrastructure and transcript-capture pipelines for autonomous multi-agent research system developed with the UK AI Security Institute.
- Implementing LLM monitors to detect auto-research failure modes from transcript analysis, hardening the auto-research harness.

Scale AI

San Francisco, USA

AI Engineer — Agentic Capabilities

July 2025 – April 2026

- Building multi-agent research systems.

Two Sigma

New York, NY, USA

Software Engineer — Platform Engineering

August 2022 – July 2025

- Workflow Orchestration Platform, Distributed systems, Centralized database service, Tech Risk/Authorization/Security.

Amazon

Seattle, WA, USA

Software Development Engineer

April 2020 – July 2022

EDUCATION

University of Southern California

2013 – 2019

M.S., Computer Science (2019); B.S., Mechanical Engineering (2017)

SKILLS

Python, PyTorch, TransformerLens, Agentic AI systems, RAG, Temporal, AWS, GCP, Kubernetes, Linux/Unix, PostgreSQL